VARIABILI:

x_1: QUANTITÁ PRODOTTO A ESPR. IN TONNELLATE
x_2: " " B " "

VINCOLI:

- $3x_1 + 2x_2 \leq 17$
- $x_2 \leq 6$
- $x_1 \geq 1$

LP:

$$\max z: \quad 4x_1 + x_2$$
$$\begin{aligned} 3x_1 + 2x_2 &\leq 17 \\ x_2 &\leq 6 \\ x_1 &\geq 1 \\ x_1,\ x_2 &\geq 0 \end{aligned}$$

FORMA STANDARD:

$$\min(-z): \quad -4x_1 - x_2$$
$$\begin{aligned} 3x_1 + 2x_2 + x_3 &= 17 \\ x_2 + x_4 &= 6 \\ x_1 - x_5 &= 1 \\ x_1, x_2, x_3, x_4, x_5 &\geq 0 \end{aligned}$$

0	-4	-1	0	0	0
17	3	2	1	0	0
6	0	1	0	1	0
1	1	0	0	0	-1

NO BASE.

INTRODUCO x^a_1 e MINIMIZZO

$\zeta_1 = \sum_{i=1} x^a_i = x^a_1$

0	0	0	0	0	0	1
17	3	2	1	0	0	0

	6	0	1	0	1	0	0
	1	1	0	0	0	-1	1

$-\zeta_1$	-1	-1	0	0	0	1	0
$x_3 =$	17	3	2	1	0	0	0
$x_4 =$	6	0	1	0	1	0	0
$x_1^a =$	1	1	0	0	0	-1	1

$B = \{A_3, A_4, A_6\}$

$x = (0, 0, 17, 6, 0, 1)$

$\alpha = (0, 0)$

$s = 1$

$$\theta_{MAX} = \min_{i : y_{i1} > 0} \frac{y_{i0}}{y_{i1}} = \min\left\{\frac{17}{3}, \frac{1}{1}\right\} = 1$$

$-\zeta_1$	0	0	0	0	0	0	1
$x_3 =$	14	0	2	1	0	3	-3
$x_4 =$	6	0	1	0	1	0	0
$x_1 =$	1	1	0	0	0	-1	1

$B = \{A_3, A_4, A_1\}$

$x = (1, 0, 14, 6, 0, 0)$

$\beta = (1, 0)$

REINTRODUCO IL VETT. COSTI INIZ.

0	-4	-1	0	0	0	0

	14	0	2	1	0	3	-3
	6	0	1	0	1	0	0
	1	1	0	0	0	-1	1

z	4	0	-1	0	0	-4	4
$x_3 =$	14	0	2	1	0	3	-3
$x_4 =$	6	0	1	0	1	0	0
$x_1 =$	1	1	0	0	0	-1	1

$B = \{A_3, A_4, A_1\}$

$x = (1, 0, 14, 6, 0, 0)$

REGOLA DI DANTZIG $\rightarrow$ $s = 5$

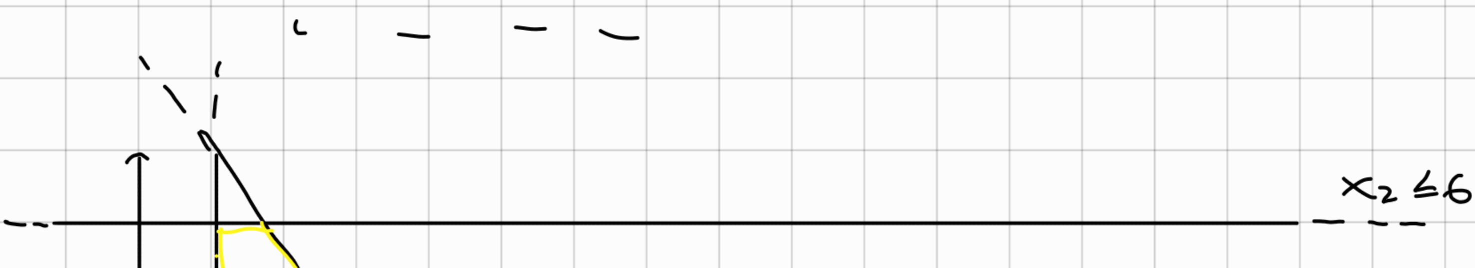
$$\theta_{MAX} = \min_{i : y_{is} > 0} \frac{y_{i0}}{y_{is}} = \min \left\{ \frac{14}{3} \right\} = \frac{14}{3}$$

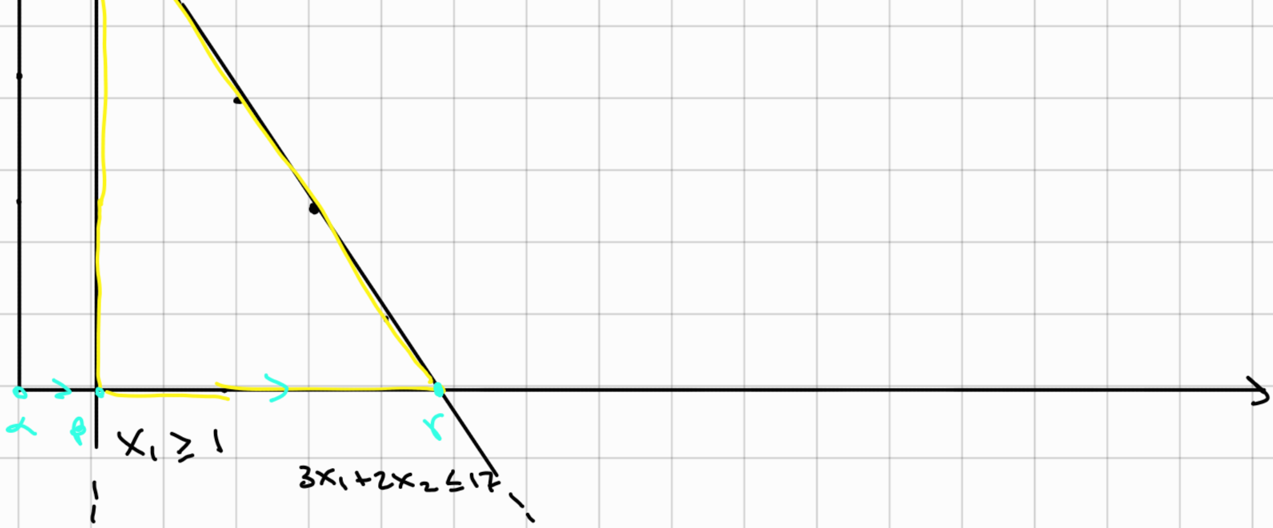
z	$\frac{68}{3}$	0	$\frac{5}{3}$	$\frac{4}{3}$	0	0	0
$x_5 =$	$\frac{14}{3}$	0	$\frac{2}{3}$	$\frac{1}{3}$	0	1	-1
$x_4 =$	6	0	1	0	1	0	0
$x_1 =$	$\frac{17}{3}$	1	$\frac{2}{3}$	$\frac{1}{3}$	0	0	0

$B = \{A_5, A_4, A_1\}$

$x = \left(\frac{17}{3}, 0, 0, 6, \frac{14}{3}, 0\right)$

$\bar{x} = \left(\frac{17}{3}, 0\right)$





$$\min(-z): -4x_1 - x_2$$

$$\begin{aligned} 3x_1 + 2x_2 + x_3 \quad &= 17 \\ x_2 \quad + x_4 &= 6 \\ x_1 \quad - x_5 &= 1 \\ x_1, x_2, x_3, x_4, x_5 &\geq 0 \end{aligned}$$

DUALE:

$$\max w: 17\pi_1 + 6\pi_2 + \pi_3$$

$$\begin{aligned} 3\pi_1 \quad + \pi_3 &\leq -4 \\ 2\pi_1 + \pi_2 &\leq -1 \\ \pi_1 &\leq 0 \\ \pi_2 &\leq 0 \\ -\pi_3 &\leq 0 \\ \pi_1, \pi_2, \pi_3 &\gtrless 0 \end{aligned}$$

LA SOL. OTTIMA DUALE π_B SI PUÓ CALCOLARE ANALIZZANDO I COSTI RELATIVI DEI TABLEAU INIZIALE e FINALE:

c_B, I COSTI IN BASE DEL TAB. INIZ. SONO TUTTI NULLI.

$\bar{c}_B$, I COSTI RELATIVI DEL TAB. FINALE CORRISPONDENTI ALLE COLONNE DELLA BASE INIZIALE $B = \{A_3, A_4, A_6\}$ SONO $\left(\frac{4}{3}, 0, 0\right)$

$$\pi_B = c_B - \bar{c}_B = \left(-\frac{4}{3}, 0, 0\right)$$

ILP!

$\max z: \quad 4x_1 + x_2$

$3x_1 + 2x_2 \leq 17$

$x_2 \leq 6$

$x_1 \geq 1$

$x_1, \quad x_2 \geq 0$

$x_1, x_2 \in \mathbb{Z}$

TAB. FINALE SENZA VAR ART.

z	$\frac{68}{3}$	0	$\frac{5}{3}$	$\frac{4}{3}$	0	0
$x_5 =$	$\frac{14}{3}$	0	$\frac{2}{3}$	$\frac{1}{3}$	0	1
$x_4 =$	6	0	1	0	1	0
$x_1 =$	$\frac{17}{3}$	1	$\frac{2}{3}$	$\frac{1}{3}$	0	0

GOMORY RIGA GEN. 3

$$\frac{2}{3}x_2 + \frac{1}{3}x_3 \geq \left(\frac{17}{3} - 5\right)$$

$$\frac{2}{3}x_2 + \frac{1}{3}x_3 \geq \frac{2}{3}$$

$$\{ x_3 = 17 - 3x_1 - 2x_2$$

$$\cancel{\frac{2}{3}x_2} + \frac{17}{3} - x_1 - \cancel{\frac{2}{3}x_2} \geq \frac{2}{3}$$

$$-x_1 \geq \frac{2}{3} - \frac{17}{3}$$

→ $x_1 \le 5$

TAGLIO IN FORMA STANDARD:

$$-\frac{2}{3}x_2 - \frac{1}{3}x_3 + x_6 = -\frac{2}{3}$$

z	$\frac{68}{3}$	0	$\frac{5}{3}$	$\frac{4}{3}$	0	0	0
$x_5 =$	$\frac{14}{3}$	0	$\frac{2}{3}$	$\frac{1}{3}$	0	1	0
$x_4 =$	6	0	1	0	1	0	0
$x_1 =$	$\frac{17}{3}$	1	$\frac{2}{3}$	$\frac{1}{3}$	0	0	0
$x_6 =$	$-\frac{2}{3}$	0	$-\frac{2}{3}$	$-\frac{1}{3}$	0	0	1

$B = \{A_5, A_4, A_1, A_6\}$

AMMISSIBILE SOLO PER DUALE → SIMPL. DUALE

$y_{40} < 0 \Rightarrow i = 4$

PIVOT:

$$\max_{j: y_{4j}<0} \frac{y_{0j}}{y_{4j}} = \max\left\{\frac{5}{3}\left(-\frac{3}{2}\right), \frac{4}{3}(-3)\right\} = \max\{-2{,}5, -4\} = -2{,}5$$

z	21	0	0	$\frac{1}{2}$	0	0	$\frac{5}{2}$
$x_5 =$	4	0	0	0	0	1	1
$x_4 =$	5	0	0	$-\frac{1}{2}$	1	0	$-\frac{3}{2}$
$x_1 =$	5	1	0	0	0	0	1

$B = \{A_5, A_4, A_1, A_2\}$

$x = (5, 1, 0, 5, 4, 0)$

$\delta = (5, 1)$

AMMISSIBILE PER

$x_2 =$ 1	0	1	$\frac{1}{2}$	0	0	$-\frac{3}{2}$

PRIMALE